Zachary Bretton

zhbretton@gmail.com • + 1 (917) 224-7086 • linkedin.com/in/zachary-bretton

SUMMARY

PhD in Cognitive and Computational Neuroscience with 6+ years of experience applying machine learning, statistical modeling, and scalable data pipelines to high-dimensional datasets. Proficient in Python, SQL, and HPC frameworks (Docker, Dask, PySpark), with expertise in predictive modeling, feature engineering, and big data analytics. Proven ability to translate complex analyses into actionable insights, lead cross-functional teams, and deliver reproducible, production-ready workflows

EDUCATION

UNIVERSITY OF TEXAS AT AUSTIN — **Austin, TX**
Ph.D. in Neuroscience — **2019 - 2024**
Thesis: The neural mechanisms and long-term impacts of working memory suppression

BOSTON UNIVERSITY — **Boston, MA**
B.A. Biology, Specialization in Neurobiology — **2011 - 2015**

PROFESSIONAL EXPERIENCE

DCS CORP — **Dayton, OH**
Cognitive Neuroscientist (Research Scientist) — **October 2024 - Present**

- Working with the Air Force Research Labs' Non-Invasive Brain Stimulation team to advance neuroimaging research on cognitive performance and brain stimulation.
- Led the development of scalable data pipelines in Python and SQL for high-volume neuroimaging data, improving computational efficiency and reproducibility across projects.
- Built and validated machine learning models (classification, connectivity, pattern recognition) to extract insights from complex neuroimaging datasets and evaluate intervention outcomes.
- Reduced analysis runtime by 90% using distributed computing (Dask, Docker), enabling rapid iteration and insights.

UNIVERSITY OF TEXAS AT AUSTIN — **Austin, TX**
Graduate Research Assistant, Lewis-Peacock Lab — **August 2019 - August 2024**

- Developed and optimized predictive ML models with scikit-learn, improving classification accuracy of high-dimensional time-series datasets through feature engineering and hyperparameter tuning, leading to multiple publications.
- Built end-to-end pipelines in Python and MATLAB for preprocessing, cleaning, and normalizing large datasets, automating repetitive tasks and improving research throughput.
- Designed and implemented rigorous experimental protocols for fMRI studies, ensuring high data integrity and reproducibility standards.
- Executed large-scale data processing on HPC clusters with Docker, SLURM, and containerization for reproducible, scalable ML workflows.

COLUMBIA UNIVERSITY MEDICAL CENTER — **New York, NY**
Lab Manager / Research Technician, Neural Circuit Lab — **September 2016 - May 2019**

- Conducted in vivo electrophysiological experiments and analyzed large-scale time-series data using MATLAB and custom pipelines, extracting features from noisy behavioral and electrophysiological datasets.
- Engineered custom experimental apparatuses (Arduino-based, 3D-printed) and standardized protocols for data collection, boosting reliability and throughput.
- Co-led studies on Early Life Stress and Working Memory in Schizophrenia, developed a novel anxiety model in mice, and contributed to multiple publications and grant submissions.

TECHNICAL SKILLS

- Proficient in Python (NumPy, SciPy, pandas, seaborn, matplotlib, nibabel, nilearn, statsmodels), MATLAB, SQL, Unix/BASH, Git, Machine Learning (scikit-learn, TensorFlow), PySpark
- Skilled in data transformation, normalization, and feature engineering; advanced statistical analysis (linear, predictive, and multivariate modeling); data visualization (Matplotlib, Seaborn); high-performance computing and containerization (HPCC, Docker, Singularity, SLURM, HTCondor, Dask)